

[TODO: THESIS TITLE TO BE FINALIZED
— depends on final RQ selection, see
docs/proposal/research/RQ_OPTIONS.md]

[TODO: AUTHOR NAME]

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Abstract

[TODO: ABSTRACT TO BE WRITTEN AFTER
RQ/ARTIFACT/METHODOLOGY ARE FINALIZED]

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Chapter 1

Introduction

1.1 Background

[TODO: BACKGROUND NARRATIVE REQUIRED — draft from PROBLEM_ARCHITECTURE.md §Context/§Phenomenon only after framing is confirmed with supervisor]

1.2 Research Context

[TODO: RESEARCH CONTEXT REQUIRED — dataset origin (Pesta Babi film frames), labeling process, and current empirical state (HSEmotion N=227 historical baseline; ArcFace+LR vs. HSEmotion N=135 common-subset comparison) to be written with denominators kept explicit and never merged]

1.3 Problem Identification

[TODO: PROBLEM STATEMENT NOT YET FINALIZED — do not draft a final Problem Statement here. See PROBLEM_ARCHITECTURE.md §1–§2 for the solution-neutral practical/knowledge/design problem chain currently on record. ArcFace must NOT be used to define the problem (guideline neutrality principle, BAB 4 §4.6).]

1.4 Research Evidence

[TODO: EVIDENCE SUMMARY REQUIRED — restate only from PROBLEM_ARCHITECTURE.md §“Claims We Can Make Now” and RESEARCH_FINDINGS.md §9. Preserve N=227 (historical) vs. N=135 (controlled common-subset) separation. Do not restate any item from §“Claims

We Cannot Make Yet.”]

1.5 Research Gap

[TODO: RESEARCH GAP NOT YET FINALIZED — four non-final gap-statement options exist in STATE_OF_ART_AND_GAP.md §9 (Options 1–4). None is selected. Do not present any option here as settled.]

1.6 Research Questions

[TODO: OFFICIAL RQ REQUIRED — RQ_OPTIONS.md §2 lists 5 non-final candidate RQs (RQ-1 through RQ-5) and a recommended working direction (Structure A+B, per the decision-support memo). No official RQ has been selected. This section must not be written until the supervisor/user makes this decision (RQ_OPTIONS.md §“Decision Required”).]

1.7 Research Objectives

[TODO: RESEARCH OBJECTIVES REQUIRED — depend directly on the RQ selected above; cannot be finalized first. See RQ_OPTIONS.md §7 for candidate (non-final) objective derivations per RQ.]

1.8 Research Scope

[TODO: SCOPE STATEMENT REQUIRED — must explicitly state the current 5-class evaluation scope (Neutral, Happy, Sad, Surprise, Fear) and the Angry/Disgust ground-truth scarcity (Angry=1, Disgust=2 across every population stage checked: 227/144/138 — see docs/R7_5_CLASS_DECISION.md) as an explicit scope boundary, not a silently omitted detail.]

1.9 Expected Contributions

[TODO: EXPECTED CONTRIBUTION REQUIRED — depends on final RQ. Do not claim design contribution by default; RQ_OPTIONS.md §8 shows several candidate RQs are primarily empirical, not design, contributions. Novelty is NOT claimed for ArcFace-as-feature-extractor (confirmed not novel, STATE_OF_ART_AND_GAP.md §7).]

1.10 Proposal Organization

This proposal is organized as follows. Chapter 2 reviews related literature and establishes the state of the art. Chapter 3 presents the intended research methodology. Chapter 4 presents the proposed design requirements and artifact (pending finalization). Chapter 5 presents the evaluation plan. **[TODO: Confirm this organization matches the final chapter set once the official RQ/artifact are locked.]**

Chapter 2

Literature Review

2.1 Facial Expression Recognition

Deep learning-based facial expression recognition (FER) is a mature, actively surveyed research area [5]. **[TODO: EXPAND — general FER background, drawing only from STATE_OF_ART_AND_GAP.md §2A. Do not add claims beyond what that document supports.]**

2.2 FER Datasets and Benchmarks

AffectNet [6] is among the largest in-the-wild FER benchmark datasets. **[TODO: EXPAND — RAF-DB and FER2013 discussion requires resolving TODO citations (see §“Citations Requiring Verification” below) before inclusion. Do not cite RAF-DB’s primary source until verified.]**

2.3 Face Representation Learning

[TODO: EXPAND — general face-representation-learning background (identity-oriented embeddings, margin-based losses) from STATE_OF_ART_AND_GAP.md §2B only.]

2.4 ArcFace and Face Recognition Representations

ArcFace [4] proposes an additive angular margin loss for face *verification*, evaluated on face-verification benchmarks; it was not designed or evaluated for expression recognition.

Important — novelty status (must not be silently omitted or softened): per the targeted literature review in docs/proposal/research/STATE_OF_ART_AND_GAP.md §7, using ArcFace-family embeddings as a frozen feature extractor for FER, followed by

a separate classifier, **has already been published** and is **not** treated as novel in this proposal.

2.5 Transfer / Downstream Use of Face Representations

At least two directly relevant precedents exist for using ArcFace-family embeddings as a frozen or near-frozen feature extractor for FER via a downstream classifier: Altaha et al. [1] (ArcFace features with a TinySiamese classifier, evaluated on FER2013, RAF-DB, and ExpW) and Waldner and Mitra [7] (ArcFace among other backbones via transfer learning, evaluated on AffectNet). **[TODO: EXPAND — explicitly state, per STATE_OF_ART_AND_GAP.md §7, that neither precedent evaluates on a skin-tone-stratified or dark-skinned-specific population – this is the candidate context-level gap, not a technique-level one.]**

2.6 FER Error Analysis

Chhua et al. [3] document that, for at least one class of system (large multimodal foundation models, evaluated on the RADIATE dataset), class-specific FER confusion is demographically stratified – e.g., Anger is misclassified as Disgust 2.1 times more often in Black female subjects than in White female subjects (for GPT-4o). **[TODO: EXPAND — this is a structurally similar but not identical phenomenon (different confusion pair, different system class) to any pattern in this thesis’s own data. Do not present it as external confirmation of this thesis’s specific Neutral→Angry finding; that finding is thesis evidence only (Chapter 3 / Chapter 5), not literature evidence.]**

2.7 Skin Tone and Demographic Robustness

Buolamwini and Gebru [2] found that commercial gender-classification systems showed substantially higher error rates for darker-skinned female subjects than for lighter-skinned male subjects, and that existing facial-analysis benchmark datasets were predominantly lighter-skinned.

Important — concept separation (must be preserved verbatim in any future draft): skin tone, race, ethnicity, and demographic representation are four distinct concepts and must not be collapsed into one another (docs/proposal/research/STATE_OF_ART_AND_GAP.md §8). This thesis’s dataset contains no race or ethnicity field. Any skin-tone finding discussed in this proposal is an

image-derived measurement, not a race or ethnicity classification, and any accuracy difference reported across skin-tone categories is an *association*, not a causal or biological claim.

[TODO: EXPAND — cite Fitzpatrick-scale critique regarding conflation with race/ethnicity once the primary Fitzpatrick source is verified (see TODO list below); until then, reference only the corroborating discussion already present in STATE_OF_ART_AND_GAP.md §2C.]

2.8 Existing Solutions

[TODO: EXPAND — summarize the 4 existing-solution categories from STATE_OF_ART_AND_GAP.md §4 (general-purpose deep FER classifiers; face-recognition-embedding-based FER; domain-adaptation methods for cross-dataset FER bias; fairness-auditing methodology). State each solution’s limitation as documented there – do not add limitations not stated in that document.]

2.9 State of the Art

[TODO: INSERT STATE-OF-THE-ART COMPARISON TABLE – see tables/state_of_art_comparison.tex placeholder. Populate directly from STATE_OF_ART_AND_GAP.md §3; do not add rows not present there.]

2.10 Research Gap

[TODO: RESEARCH GAP NOT YET FINALIZED – STATE_OF_ART_AND_GAP.md §5 lists four candidate gaps (G1 moderate, G2 moderate-to-strong, G3 moderate/contingent, G4 unsupported) and §9 lists four non-final gap-statement options. This section must not select or assert a final gap ahead of the official RQ decision (see Chapter 1, §1.5).]

Citations Requiring Verification

The following sources are referenced in STATE_OF_ART_AND_GAP.md but their metadata was **not** fully independently verified during Phase 3, and are therefore **not yet included** in references.bib. They must not be cited in this proposal until verified:

- [TODO: VERIFY — Hosseini, Pourramezan Fard & Mahoor, “Faces of Fairness” (arXiv:2502.11049) — publication year inconsistent (arXiv

numbering implies ~2025; fetched page metadata read 2026) — resolve before citing.]

- [TODO: VERIFY — “Racial Bias within Face Recognition: A Survey,” ACM Computing Surveys (arXiv:2305.00817) — full author list not independently confirmed.]
- [TODO: VERIFY — Fitzpatrick skin-type scale primary source (commonly cited as Fitzpatrick, T.B., 1988) — not fetched or verified in Phase 3.]
- [TODO: VERIFY — RAF-DB primary dataset paper (commonly cited as Li, Deng & Du, CVPR 2017) — title/author/venue combination not independently confirmed in Phase 3.]
- [TODO: VERIFY — general transfer-learning-for-FER literature (e.g., an MDPI “Transfer Learning for Facial Expression Recognition” paper) — title-level only, not independently fetched.]

Chapter 3

Research Methodology

3.1 Research Paradigm

[TODO: RESEARCH PARADIGM TO BE STATED — per the guideline (BAB 2), state whether this research is positioned as design research, or design research combined with an explanatory/evaluative orientation, once the RQ type (descriptive/explanatory/design-oriented/evaluative — guideline Table 8.1) is fixed by the official RQ.]

3.2 Design Research Methodology

[RESEARCH METHODOLOGY TO BE FINALIZED]
[TODO: Phase 7 of the proposal-design workflow (docs/proposal/design/METHODOLOGY_DESIGN.md) has not been executed. DSRM, DRM, and other candidate methodologies must be compared there before this section is written. Do not select a methodology preemptively.]

3.3 Conceptual Framework

[TODO: Conceptual framework requires the Theory/Knowledge phase (Phase 1 audit: “Theory” row = MISSING) to be completed first. See figures/conceptual_framework.tex placeholder.]

3.4 Research Activities

[TODO: Research activities depend on the finalized RQ and methodology. Do not list activities that presuppose a specific artifact (e.g., do not default to

“train ArcFace classifier” as an activity before Chapter 4’s artifact definition is finalized).]

3.5 Data Sources

The dataset consists of face crops drawn from the film *Pesta Babi*. [TODO: EXPAND — describe data source provenance factually, matching PROBLEM_ARCHITECTURE.md only. Note: formal data-provenance/consent documentation for this source is currently flagged as a gap (docs/proposal/validation/PROPOSAL_GUIDELINE_AUDIT.md, “Ethics” row) and must be addressed before this section is finalized.]

3.6 Dataset Preparation

[TODO: Describe the existing pipeline (face detection, cropping) factually once this chapter is finalized. Source: src/fer_dataset/pipeline/ (code, not to be modified in this phase).]

3.7 Ground Truth / Annotation

Ground truth is a manually assigned expression label per face crop (N=227 historical population). [TODO: EXPAND — cite data/1408-1010-intermediate/manual_labels_export.csv provenance and the Angry(1)/Disgust(2) scarcity limitation explicitly (docs/R7_5_CLASS_DECISION.md).]

3.8 Artifact Development

[TODO: ARTIFACT NOT YET DEFINED — see Chapter 4. Do not describe artifact development steps before the artifact itself is defined there.]

3.9 Demonstration

[TODO: Demonstration plan depends on the finalized artifact (Chapter 4) and cannot be written yet. Per the guideline (BAB 12 §12.5), demonstration is distinct from evaluation (“can it run” vs. “how well does it work”) and must not be conflated with Chapter 5.]

3.10 Evaluation

[TODO: See Chapter 5 for the evaluation plan skeleton. This section should cross-reference, not duplicate, that chapter once both are finalized.]

3.11 Analysis

[TODO: Analysis approach (statistical tests, error/skin-tone analysis) depends on the finalized RQ. Existing precedent methodology (McNemar’s test, group-aware bootstrap, Fisher’s exact test) is documented as *already-executed thesis evidence* in docs/RESEARCH_FINDINGS.md, not yet as a *proposed future methodology* — this distinction must be preserved when this section is written.]

3.12 Learning / Refinement

[TODO: Per the guideline (BAB 12 §12.2), the build-evaluate-learn cycle’s explicit “learning objective” per iteration must be stated once a methodology (DSRM/DRM/other) is selected.]

3.13 Validity Threats

[TODO: Construct validity, internal validity, external validity/transferability, and reliability (guideline BAB 13 §13.7) must each be addressed once the RQ and evaluation design are finalized. At minimum, the following threats are already known from existing evidence and must not be omitted: (1) small and uneven skin-tone subgroup sizes (Dark N=46, Medium-Dark N=32, Unknown N=57 of N=135); (2) Angry/Disgust ground-truth scarcity limiting generalization claims; (3) single-dataset, single-source-film scope limiting external validity.]

3.14 Ethical Considerations

[TODO: ETHICS SECTION REQUIRED — per docs/proposal/validation/PROPOSAL_GUIDELINE_AUDIT.md, the “Ethics” row is rated WEAK: no documentation currently exists for (a) data provenance/consent for the source film *Pesta Babi*, (b) face-crop data handling/retention policy, (c) a misuse-risk statement for the skin-tone finding.

This section must not be left as a formality — these are real, unresolved content gaps, not merely missing paperwork.]

Chapter 4

Proposed Design

4.1 Problem-to-Requirement Traceability

[TODO: TRACEABILITY MATRIX REQUIRED — see tables/requirements_traceability.tex placeholder. Cannot be populated before Design Requirements (Phase 5) exists. Guideline reference: BAB 10 §10.7, Table 10.3.]

4.2 Design Requirements

[DESIGN REQUIREMENTS TO BE FINALIZED]
[TODO: Phase 5 of the proposal-design workflow (docs/proposal/design/DESIGN_REQUIREMENTS.md) has not been executed. Requirements must be derived from the practical problem and root cause (Chapter 1), NOT backfilled from the already-built ArcFace+Logistic-Regression pipeline (docs/proposal/validation/PROPOSAL_GUIDELINE_AUDIT.md, “Requirements” row).]

4.3 Artifact Definition

[ARTIFACT TO BE DEFINED] [TODO: Phase 6 (docs/proposal/design/ARTIFACT_ARCHITECTURE.md) has not been executed. The artifact type (construct / model / method / framework / combination — guideline BAB 11 §11.1) is not yet determined. Do NOT default to “ArcFace + Logistic Regression” as the artifact — per docs/proposal/validation/PROPOSAL_GUIDELINE_AUDIT.md, this is currently only a working *instantiation*, not

a guideline-compliant artifact specification, and per docs/proposal/research/STATE_OF_ART_AND_GAP.md §7, ArcFace-as-feature-extractor for FER is not novel as a technique — any artifact contribution claim here must be scoped to context/evaluation-design, not the technique itself.]

4.4 Artifact Architecture

[TODO: See figures/artifact_architecture.tex placeholder. Cannot be populated before §4.3 is finalized.]

4.5 Design Rationale

[TODO: Per the guideline (BAB 11 §11.7), each artifact component requires a design rationale of the form: “because knowledge K shows mechanism M affects outcome O, the artifact provides function F to influence M.” This requires the Theory/Knowledge phase (currently MISSING, per docs/proposal/validation/PROPOSAL_GUIDELINE_AUDIT.md) to be completed first.]

4.6 Components

[TODO: Component breakdown depends on §4.3. Do not list components (e.g., “face detector,” “embedding extractor,” “classifier”) as if they were already-finalized design decisions rather than reused instantiation details from R5–R9.]

4.7 Assumptions

[TODO: State explicitly which conditions are presumed true (guideline BAB 4 §4.7 distinguishes assumption from constraint from limitation) once Design Requirements exist. Do not conflate with §4.9 or Chapter 5’s validity threats.]

4.8 Constraints

[TODO: At minimum, the following constraints are already known from existing evidence and must be carried into this section once written: (1) 5-class evaluation scope only (Angry=1, Disgust=2 ground-truth scarcity at every

population stage: 227/144/138 — docs/R7_5_CLASS_DECISION.md); (2) small, uneven skin-tone subgroups if a skin-tone-stratified design is pursued (Dark N=46, Medium-Dark N=32, Unknown N=57 of N=135).]

4.9 Expected Behavior

[TODO: Per Rule 4 (docs/proposal/ plan): the proposed artifact does NOT have to outperform the baseline for the thesis to be meaningful. Any expected-behavior statement here must explicitly allow for a negative/null result as a valid outcome, consistent with the already-observed R7 result (ArcFace+LR vs. HSEmotion accuracy not significantly different, McNemar $p = 0.4614$, N=135) — this existing result must be labeled as *existing preliminary evidence*, not a *future expected* result, if referenced here.]

Limitations of the Current Design State

This chapter is intentionally incomplete. Per docs/proposal/validation/PROPOSAL_GUIDELINE_AUDIT both the “Requirements” and formal “Artifact” specification rows are not yet READY. This chapter must not be completed until Phases 5 and 6 of docs/proposal/ are executed and approved.

Chapter 5

Evaluation Plan

5.1 Evaluation Objectives

[TODO: Evaluation objectives depend on the finalized RQ (Chapter 1) and artifact (Chapter 4). Do not state objectives that presuppose an artifact form not yet defined.]

5.2 Evaluation Criteria

[TODO: Per the guideline (BAB 13 §13.5): criterion → indicator → metric, each derived from Requirements/Objectives — cannot be finalized before Chapter 4’s Design Requirements exist.]

5.3 Evaluation Metrics

[TODO: Candidate metrics already demonstrated as usable in this dataset (existing evidence, not yet re-confirmed as the final proposal’s metrics): accuracy, macro precision/recall/F1, per-class error rate, false-Angry rate. Source: docs/RESEARCH_FINDINGS.md §2–§3. Final metric selection depends on the finalized RQ.]

5.4 Baseline

The existing baseline classifier already evaluated in this dataset is HSEmotion. [TODO: EXPAND — describe HSEmotion’s role as baseline classifier (not “ArcFace emotion classifier” terminology anywhere in this document — ArcFace is a frozen face-representation approach, per docs/proposal/research/RQ_OPTIONS.md §14 terminology rule).]

5.5 Experimental Design

[TODO: Experimental design (e.g., cross-validation strategy, grouping strategy) depends on the finalized methodology (Chapter 3). Existing precedent (StratifiedGroupKFold with temporal-block grouping, tools/train_arcface_classifier.py) is documented as *already-executed thesis evidence*, not yet re-adopted as the final proposal’s design.]

5.6 Error Analysis

Existing preliminary evidence (N=135 common-evaluation subset, not a future proposal result): HSEmotion’s Neutral ground-truth samples are disproportionately misclassified as Angry ($18/76 = 23.68\%$, identically reproduced on the N=227 historical population and the N=135 common subset). [TODO: EXPAND — this is reported here as already-obtained evidence motivating further characterization (per the working direction, Structure A+B), not as a result this proposal will newly produce. Any future error-analysis plan must be stated separately from this existing finding.]

5.7 Skin-Tone-Stratified Analysis

Existing preliminary evidence (N=135 common-evaluation subset): an association was observed between skin-tone category and HSEmotion accuracy (Dark $n = 46$: 45.65%; Medium-Dark $n = 32$: 12.5%; Fisher’s exact test, OR=5.88, $p = 0.0027$). The Unknown skin-tone bucket ($n = 57$ of 135) is excluded from this comparison, not imputed.

This association must not be described as causal, as evidence of racial or ethnic bias, or as Papuan-specific, consistent with docs/RESEARCH_FINDINGS.md §10 and docs/proposal/research/STATE_OF_ART_AND_GAP.md §8. The dataset contains no ethnicity field, and skin tone (an image-derived LAB-luminance bucket) is not a valid proxy for ethnicity.

[TODO: EXPAND — state whether/how this secondary analysis will be extended in the final proposal (per the working-direction recommendation, Structure A+B treats this as a secondary, not primary, analytical dimension) — pending official RQ confirmation.]

5.8 Statistical Analysis

Existing preliminary evidence: McNemar’s exact test (paired comparison, N=135: $b = 20$, $c = 26$, $p = 0.4614$ — not statistically significant); group-aware boot-

strap 95% confidence intervals (ArcFace accuracy [0.3383, 0.4809], HSEmotion accuracy [0.2692, 0.4672]); Fisher’s exact test (skin-tone subgroup, reported above). **[TODO: EXPAND — final proposal must state which statistical tests are proposed for the (not-yet-finalized) RQ, distinguishing clearly from the tests already applied to the existing R7 comparison above.]**

5.9 Validity Threats

[TODO: Cross-reference Chapter 3, §3.13. Do not duplicate — this section should address evaluation-specific validity concerns only (e.g., construct validity of the skin-tone measurement, per docs/proposal/research/STATE_OF_ART_AND_GAP.md §8’s explicit warning that this thesis’s image-derived skin-tone measurement is not equivalent to the literature’s typical race/Fitzpatrick measurement).]

5.10 Expected Evidence

[TODO: State what evidence the finalized RQ would require to be answered (per docs/proposal/research/RQ_OPTIONS.md §6–§7 for each non-final candidate RQ’s data/evidence requirements). Do not present already-obtained R1–R9 evidence as if it were yet to be produced.]

Bibliography

- [1] Mohammed A. Altaha, Islem Jarraya, Tarek M. Hamdani, and Adel M. Alimi. Facial expression recognition based on ArcFace features and TinySiamese network. In *2023 International Conference on Cyberworlds (CW)*, pages 24–31. IEEE, 2023. Confirmed via IEEE Xplore listing (document 10337533). See docs/proposal/research/STATE_OF_ART_AND_GAP.md Section 11.
- [2] Joy Buolamwini and Timnit Gebru. Gender shades: Intersectional accuracy disparities in commercial gender classification. In *Proceedings of Machine Learning Research*, volume 81, 2018. Conference on Fairness, Accountability and Transparency (FAT*). Confirmed via PMLR proceedings page. See docs/proposal/research/STATE_OF_ART_AND_GAP.md Section 11.
- [3] Kimtiya Chhua, Zhiyao Wen, Vishal Hathalia, Kai Zhu, and Shane O’Brien. From bias to balance: Detecting facial expression recognition biases in large multimodal foundation models. *arXiv preprint arXiv:2408.14842*, 2024. Confirmed via direct fetch. See docs/proposal/research/STATE_OF_ART_AND_GAP.md Section 11.
- [4] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. Confirmed via CVF Open Access (openaccess.thecvf.com) and dblp. See docs/proposal/research/STATE_OF_ART_AND_GAP.md Section 11.
- [5] Shan Li and Weihong Deng. Deep facial expression recognition: A survey. *IEEE Transactions on Affective Computing*, 2020. Confirmed via direct fetch (whdeng.cn hosted copy). See docs/proposal/research/STATE_OF_ART_AND_GAP.md Section 11.
- [6] Ali Mollahosseini, Behzad Hasani, and Mohammad H. Mahoor. AffectNet: A database for facial expression, valence, and arousal computing in the wild. *IEEE Transactions on Affective Computing*, 2017. TODO: exact publication year requires verification (2017 vs. 2019 vs. journal issue year) – flagged in docs/proposal/research/STATE_OF_ART_AND_GAP.md Section 11, item 4. Confirmed via arXiv:1708.03985 and ACM DL listing only.

- [7] David Waldner and Sunanda Mitra. Pairwise discernment of AffectNet expressions with ArcFace. *arXiv preprint arXiv:2412.01860*, 2024. Confirmed via direct fetch. See docs/proposal/research/STATE_OF_ART_AND_GAP.md Section 11.